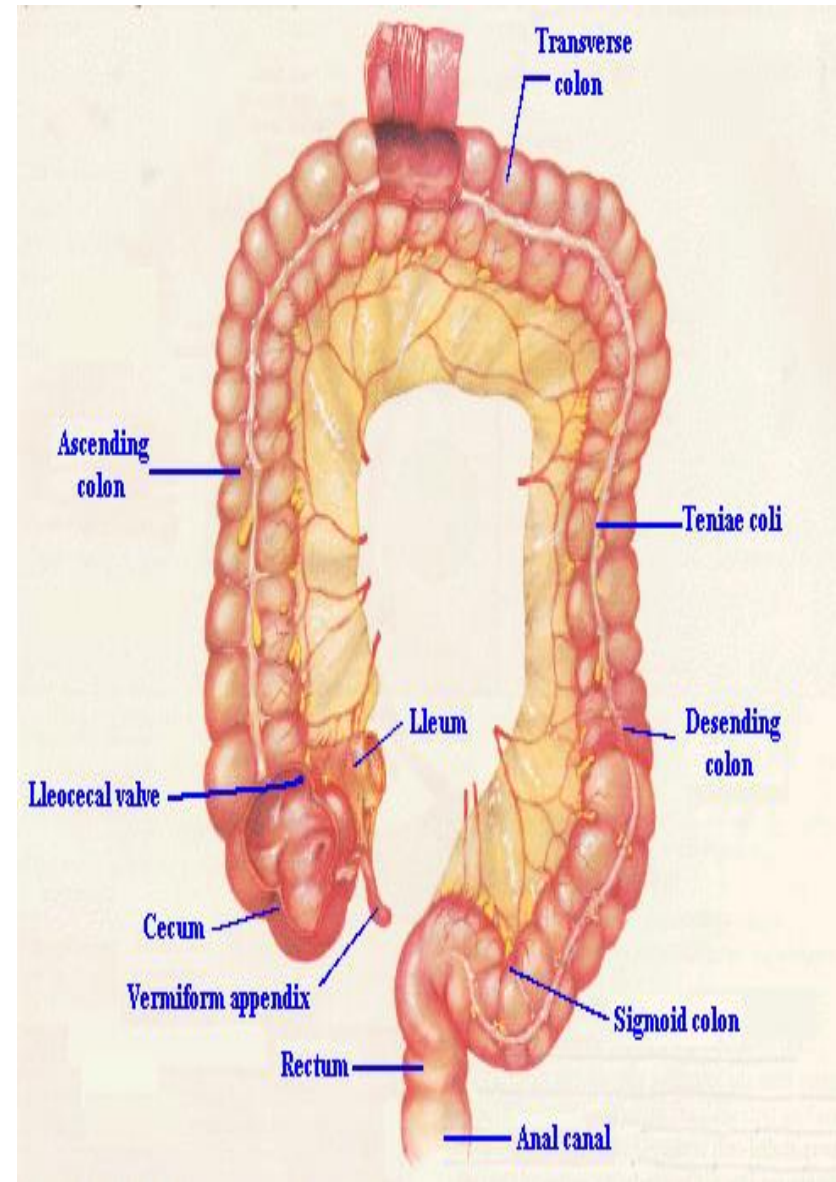


The Digestive System
Large intestine and Anal
canal
SGL 3

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Large Intestine

- Extends from ileocecal valve to anus .
- **Regions**
 - Cecum – Appendix
 - Colon(1.5 m)
 - Ascending
 - Transverse
 - Descending
 - sigmoid
 - Rectum
 - Anal canal

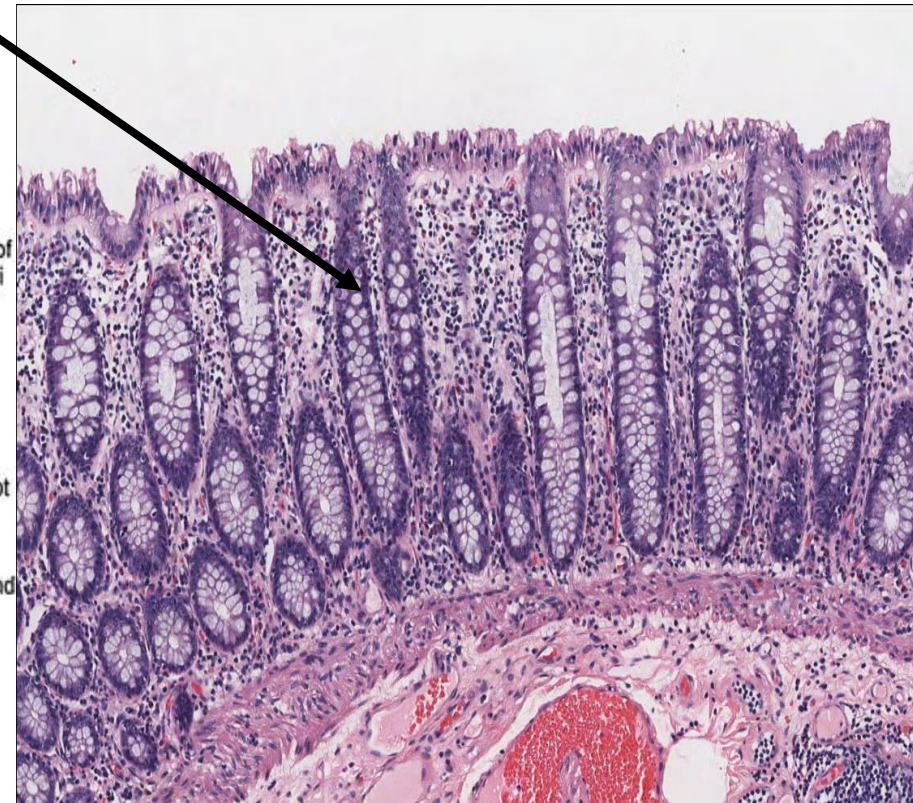
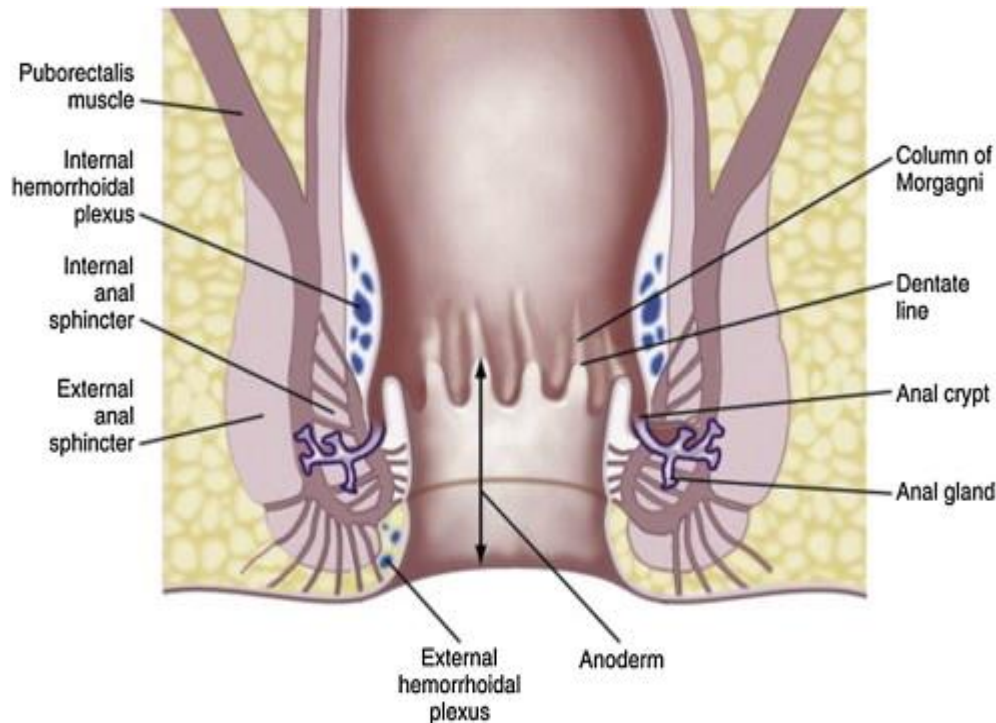


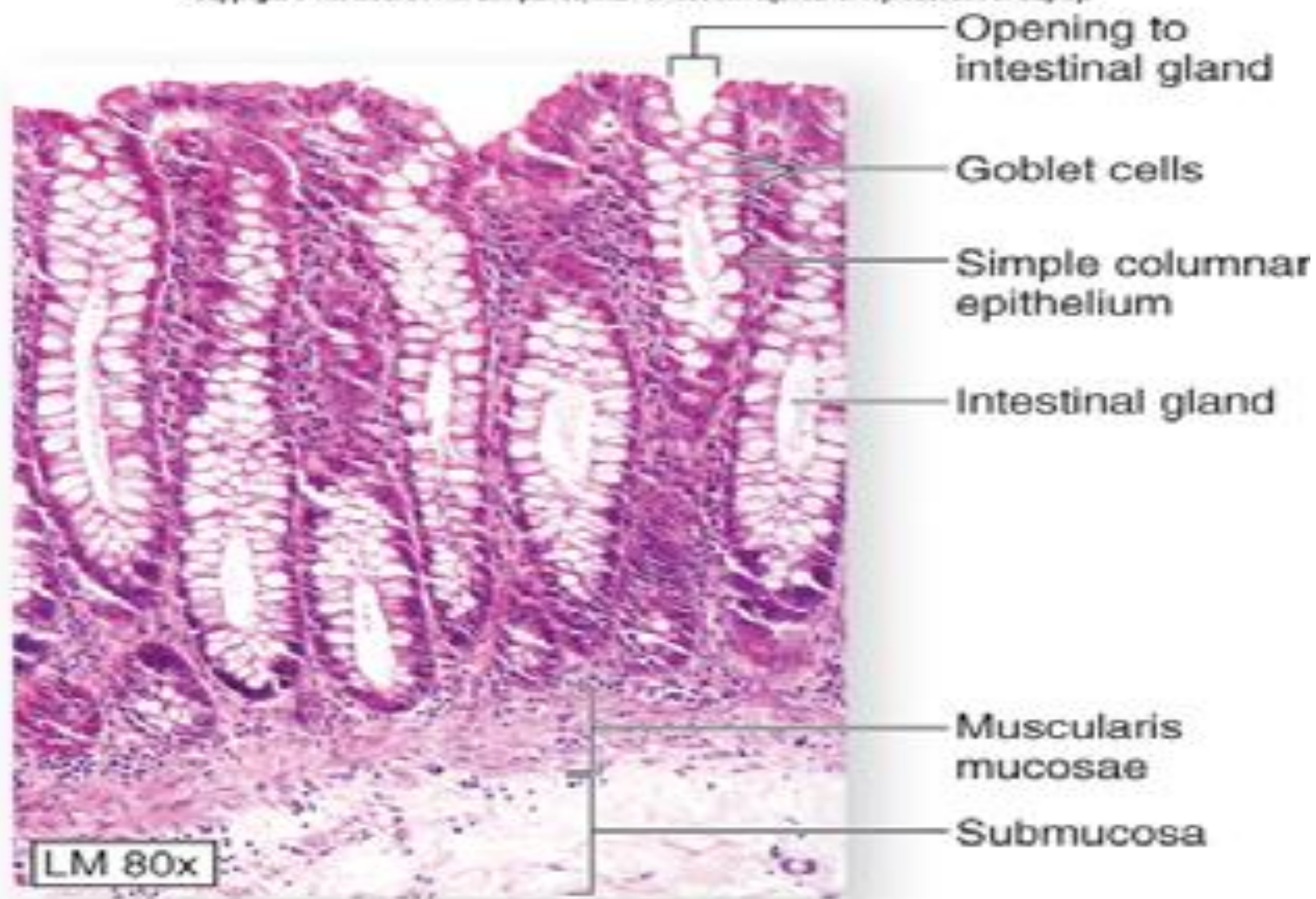
- **Histology in general:**

The large intestine is identified by:

- **Numerous goblet cells**
- **Deep crypts with no villi -no folds .**
- **it is shorter & wider lumen than small intestine**
- **Numerous lymphatic nodules**
- **Taeniae coli**
- **Crypts of Lieberkuhn long & straight**

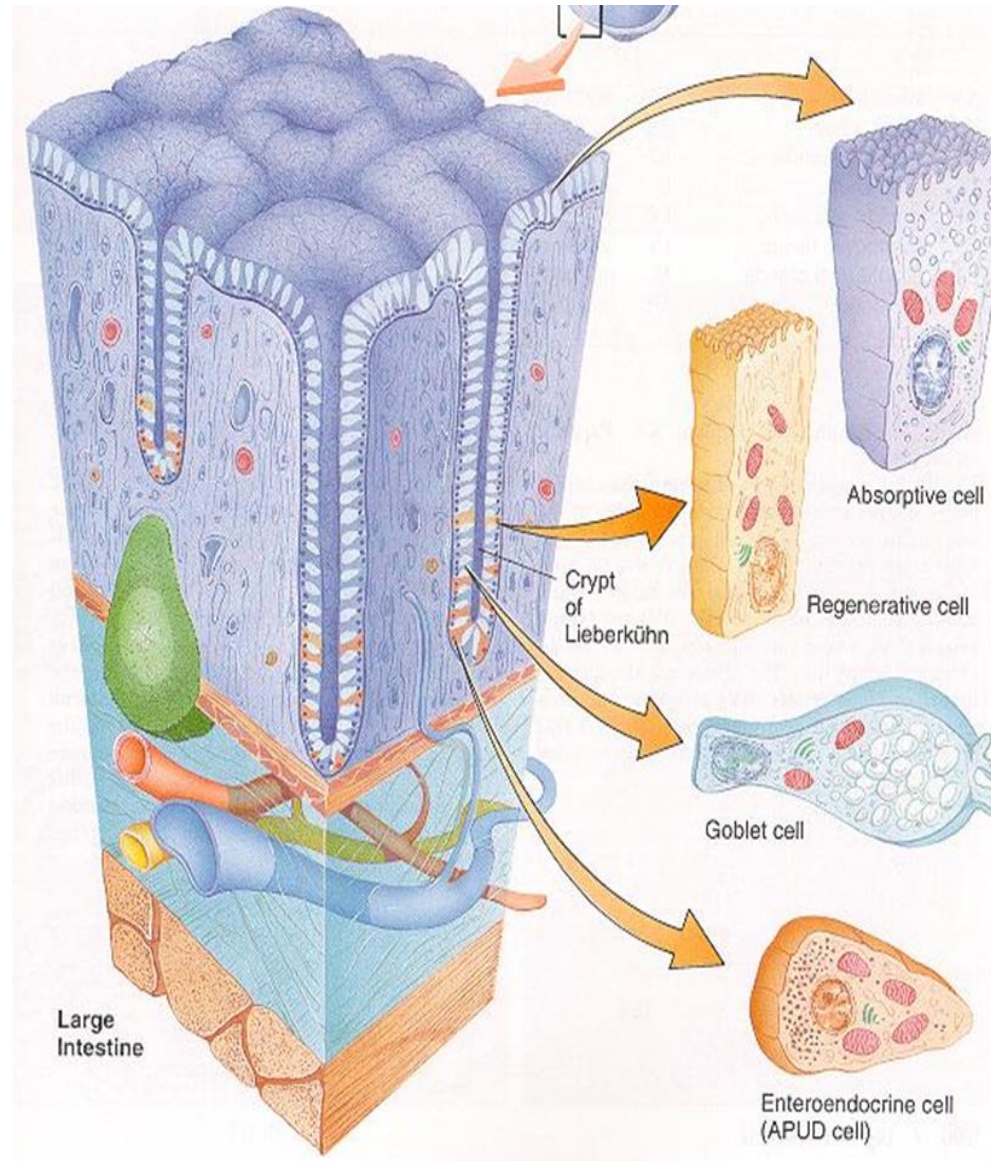
Cecum & colon The cecum is first part, from it appendix attached to it. **The wall** consist of :
A-mucosa: no fold except in rectal portion where vertical folds called **the rectal columns of morgagni** located in recto anal junction, mucosa consist of simple columnar epithelium invaginated to form simple tubular gland **crypts of lieberkuhin**,



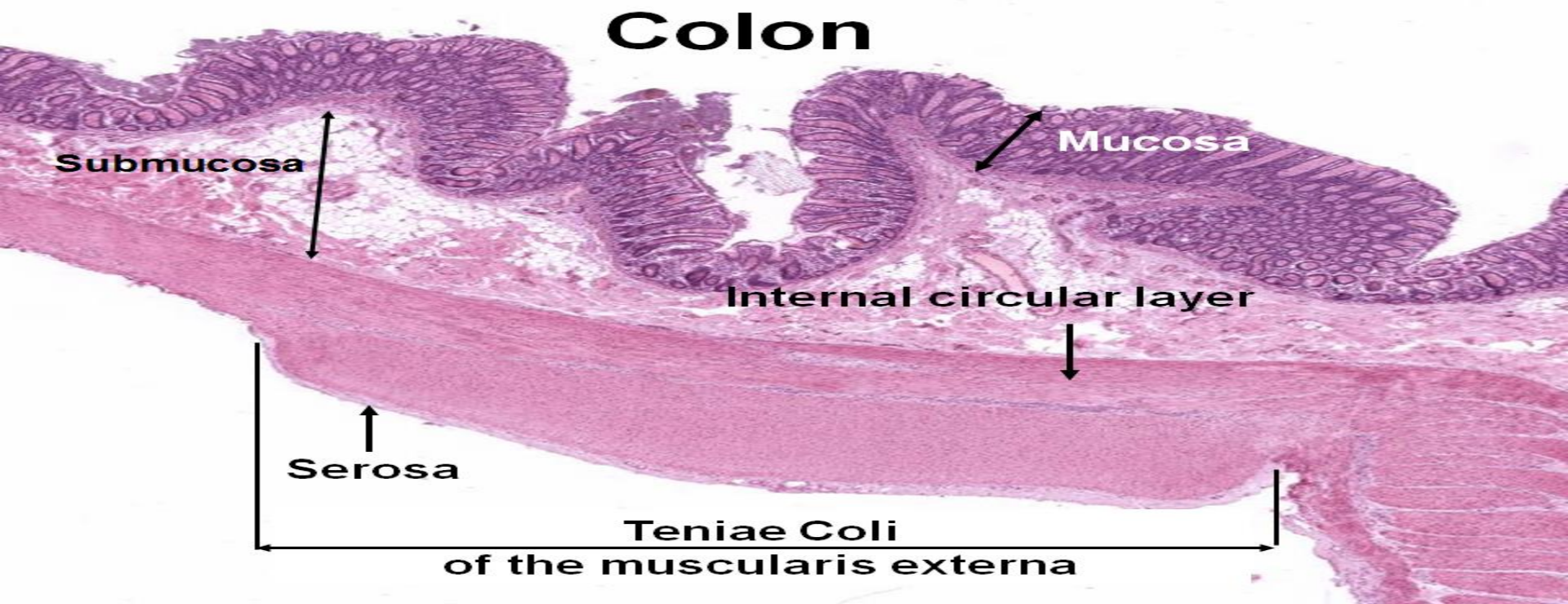


The mucosal surface & the glands are lined by an epithelium made up mainly of:

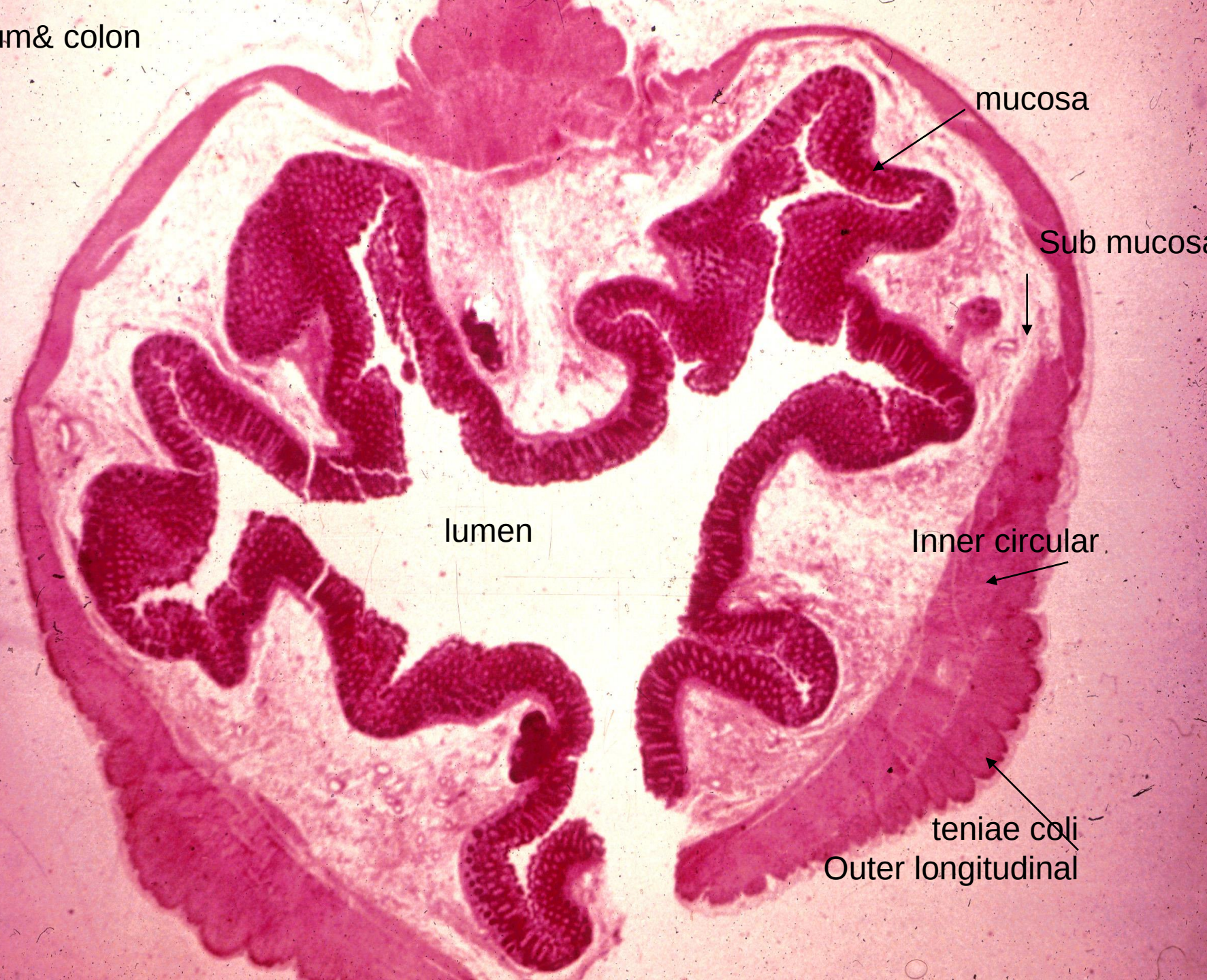
- 1-Absorptive cell (**Columnar cells**) Columnar, more numerous
- 2-Goblet cells Contain mucin granules
- For lubricant
- 3-Stem cells Base of crypts, turn over in 3 - 4 days, are precursor cells of other cells
- 4-Endocrine cells Large intestine contains some endocrine cells mainly in the lower half of crypts. Their granules found at the basal of nucleus.



- **Lamina propria** :has more lymphatic nodules than small intestine, the other layers similar to small intestine but differ from it that the outer longitudinal layer of **Muscularis externa**: in the colon is gathered into 3 thick longitudinal bands called teniae coli, with thin longitudinal fiber between band.
- **Serosa or adventitia**: **Intraperitoneal**-(caecum, transverse & sigmoid colon) –serosa
- Retroperitoneal- (ascending & descending)-adventitia & serosa
- Rectum- adventitia



Cecum & colon



mucosa

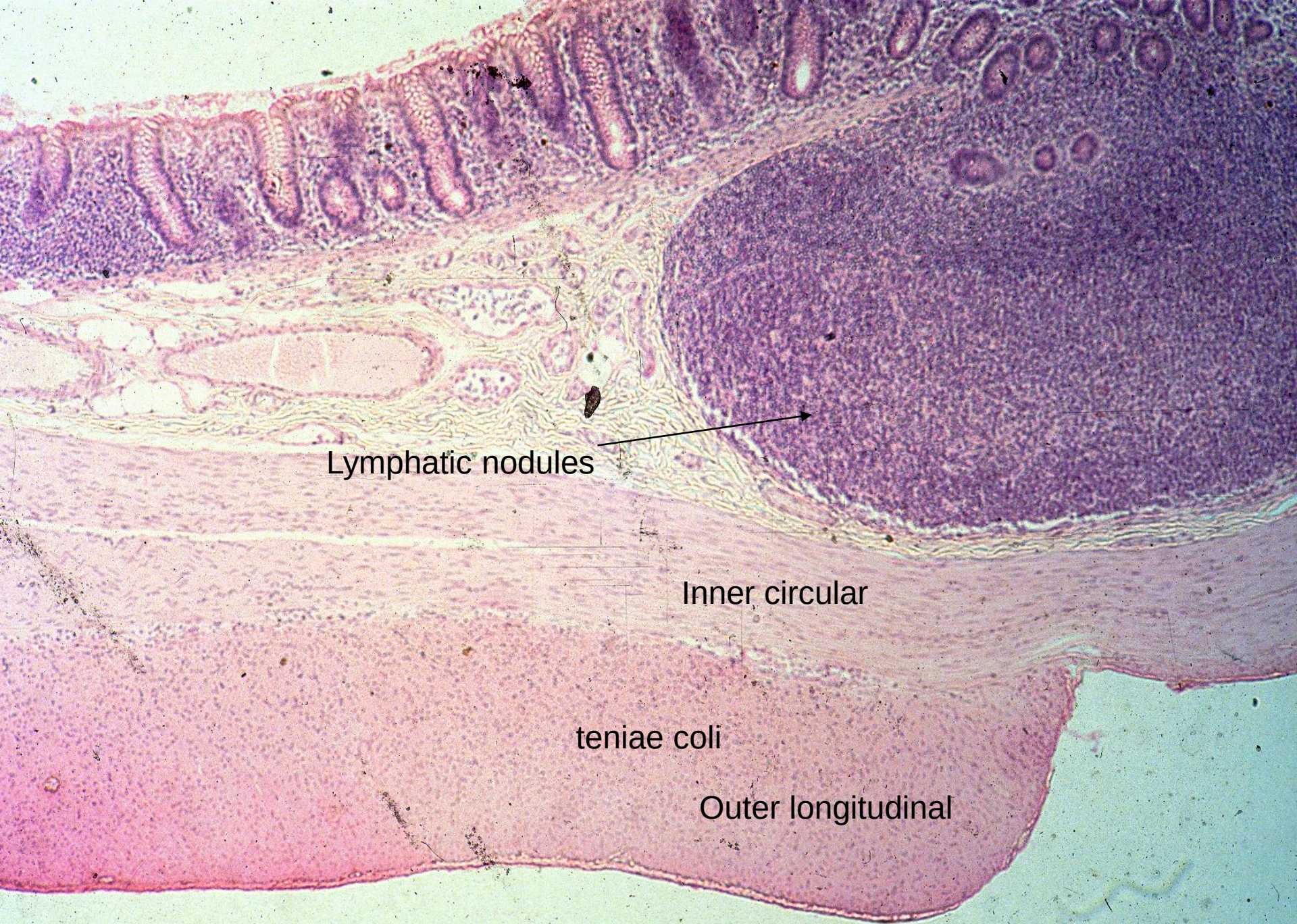
Sub mucosa

lumen

Inner circular

teniae coli

Outer longitudinal



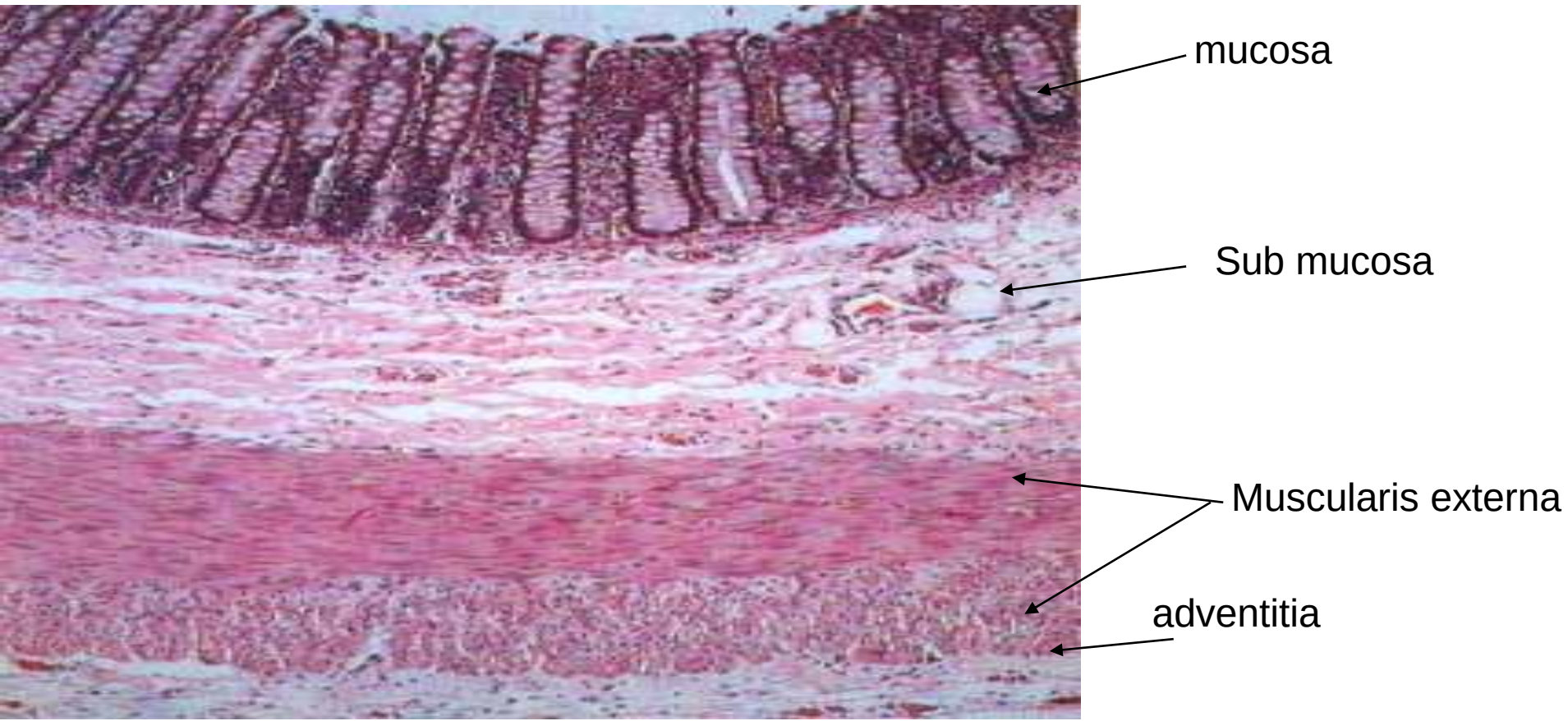
Lymphatic nodules

Inner circular

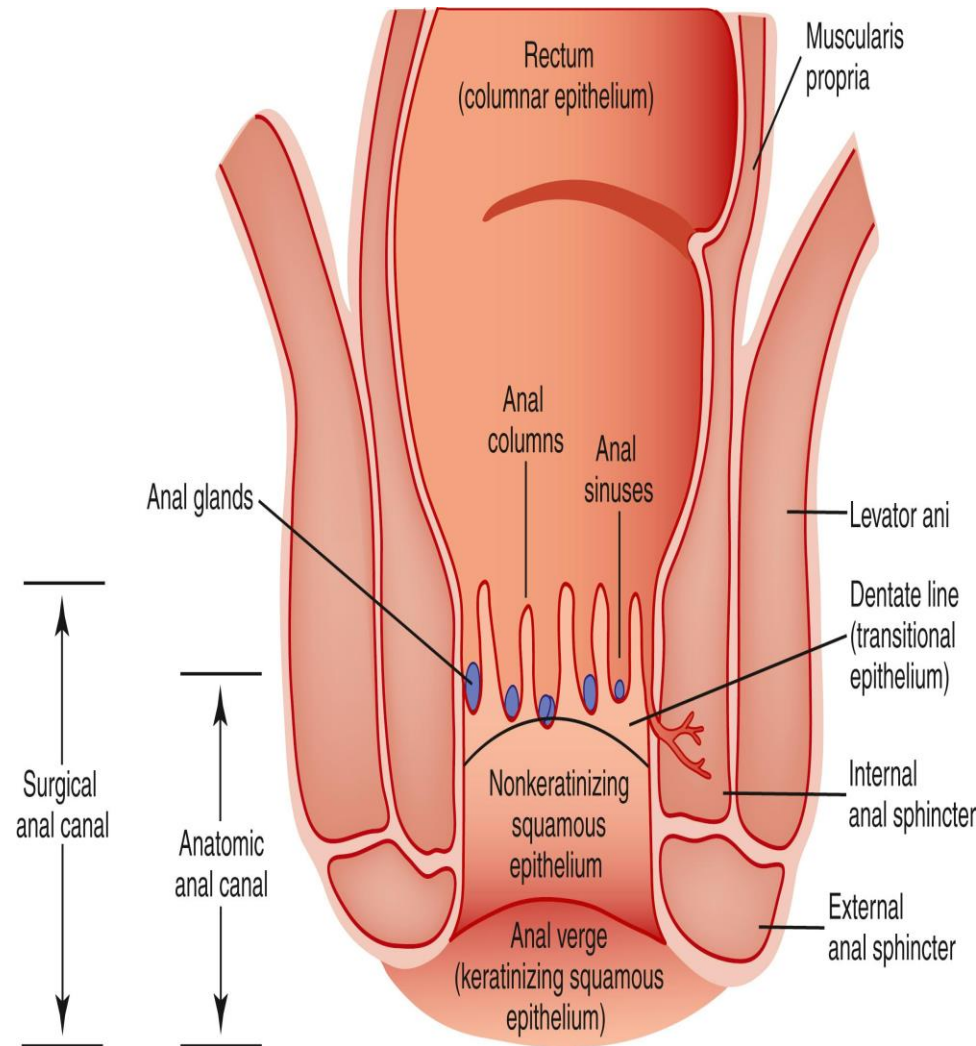
teniae coli

Outer longitudinal

- **Rectum** : The rectum's histology is similar to the rest of the large intestine, but absence of taeniae coli

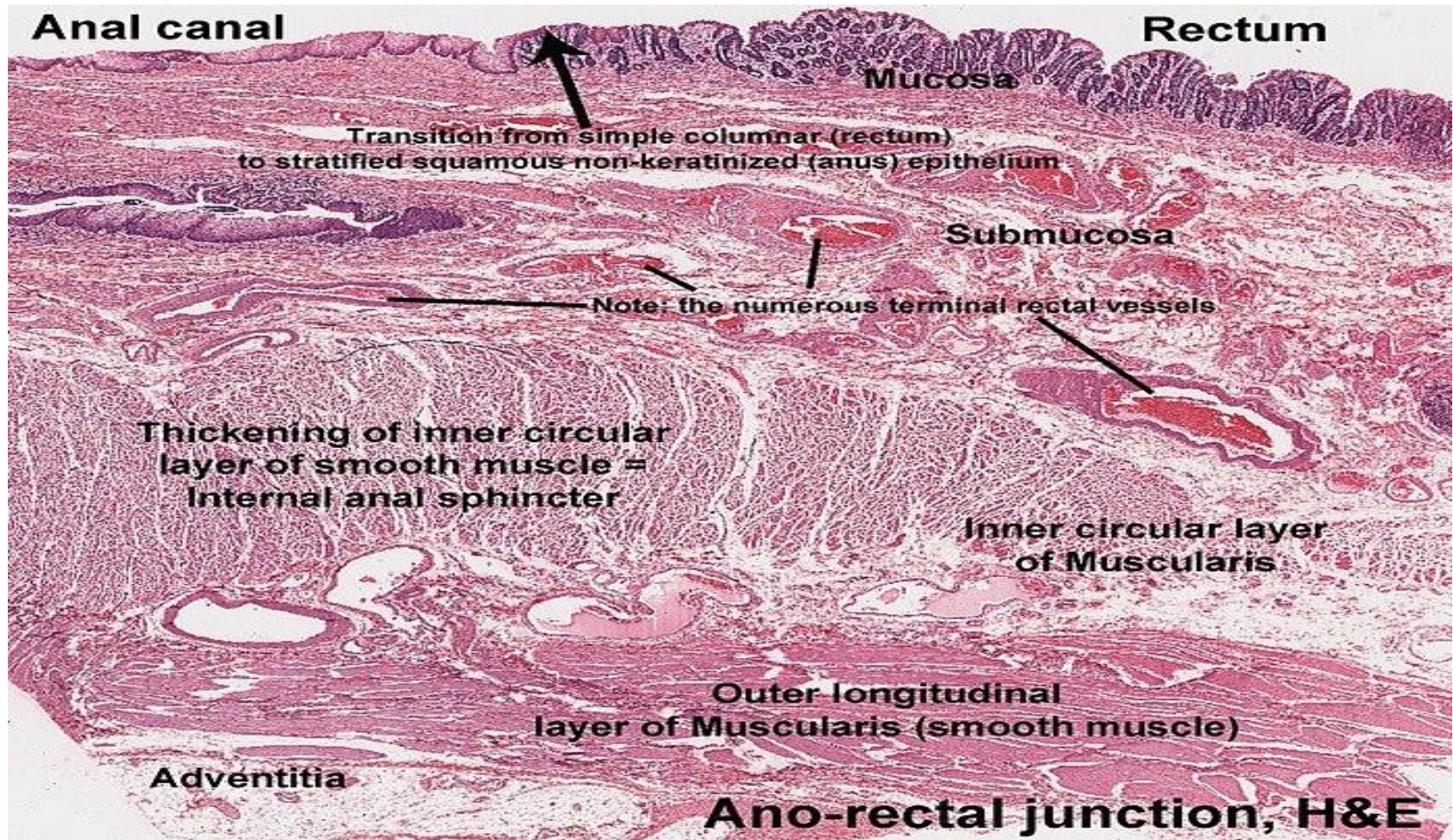


- **Anal canal :**
- About 4cm connect the rectum & the anal opening
- The upper 3 cm are lined by mucous membrane & the lower 1 cm are lined by skin.
- **The upper 3 cm** can be subdivided into 15mm upper part & 15mm lower part.
- The mucous membrane of the upper 15mm of the anal canal is lined by columnar epithelium have 6-12 longitudinal folds that are called the anal columns (Morgagni).
- The lower ends of the anal columns are united to each other by short transverse folds called the anal valves.



The anal valves form a complete ring around the anal canal, called the dentate (pectinate) line divides the anal canal into upper (two-thirds) and lower (one-third) parts,

- The mucous membrane of the lower 15 mm part is lined by non-keratinized stratified squamous epithelium.
- The lowest 8-10 mm of the anal canal are lined by true skin in which hair follicles, sebaceous glands & sweat glands are present.

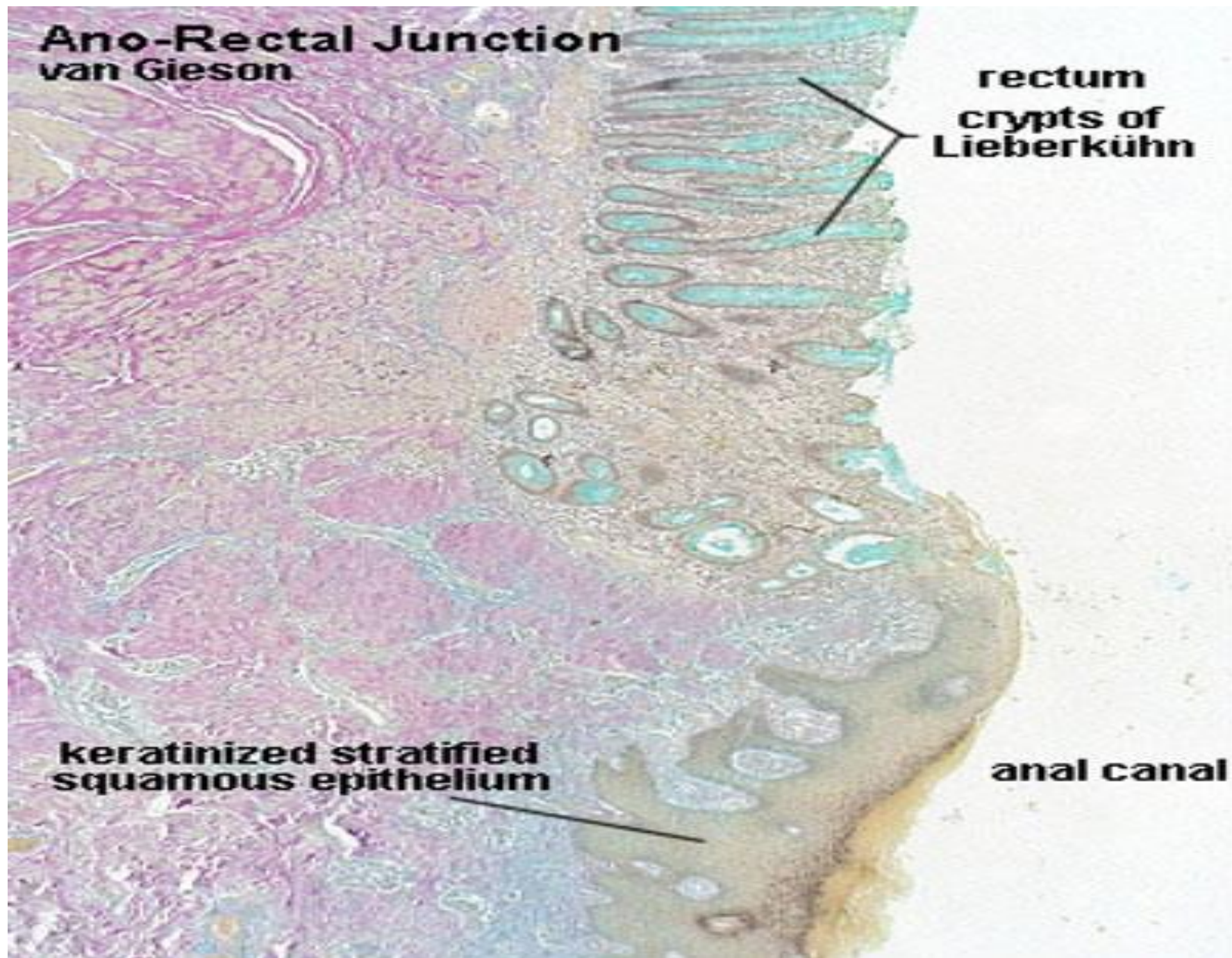


Ano-Rectal Junction van Gieson

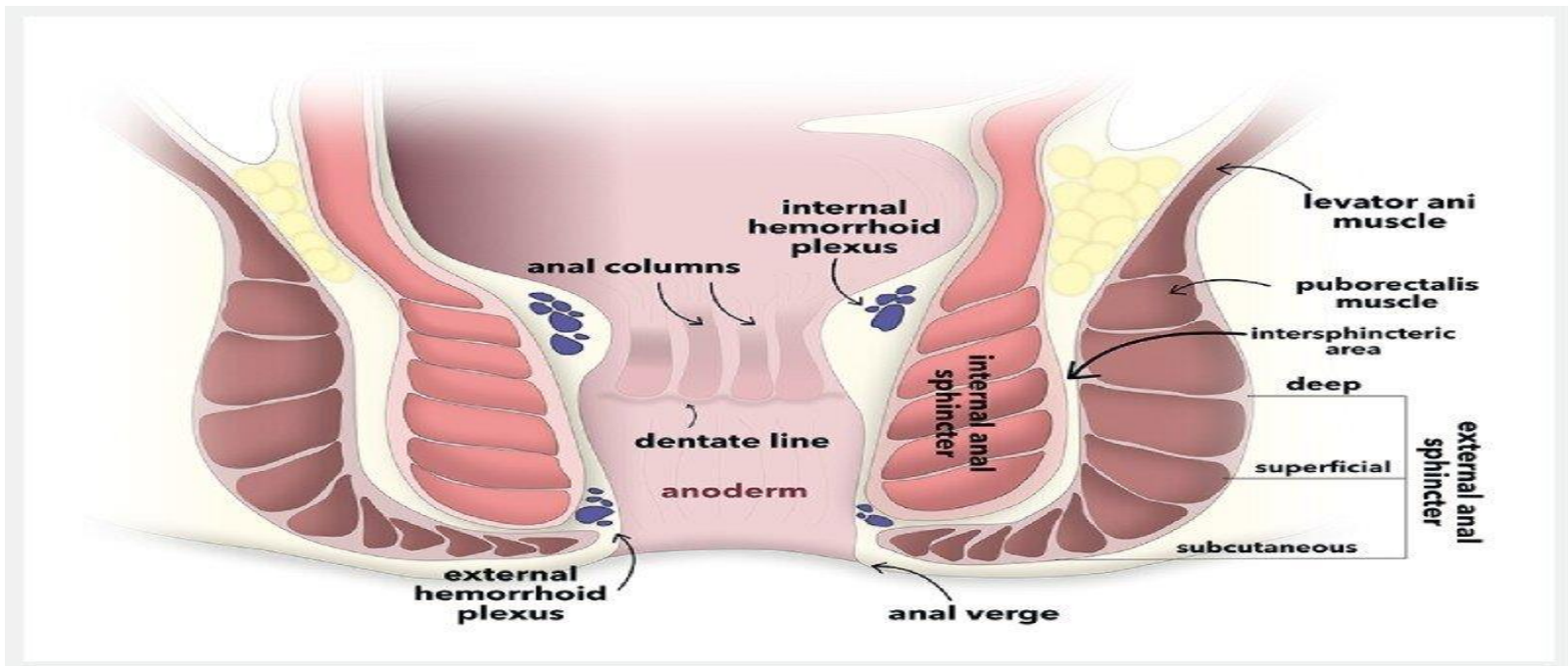
rectum
crypts of
Lieberkühn

keratinized stratified
squamous epithelium

anal canal

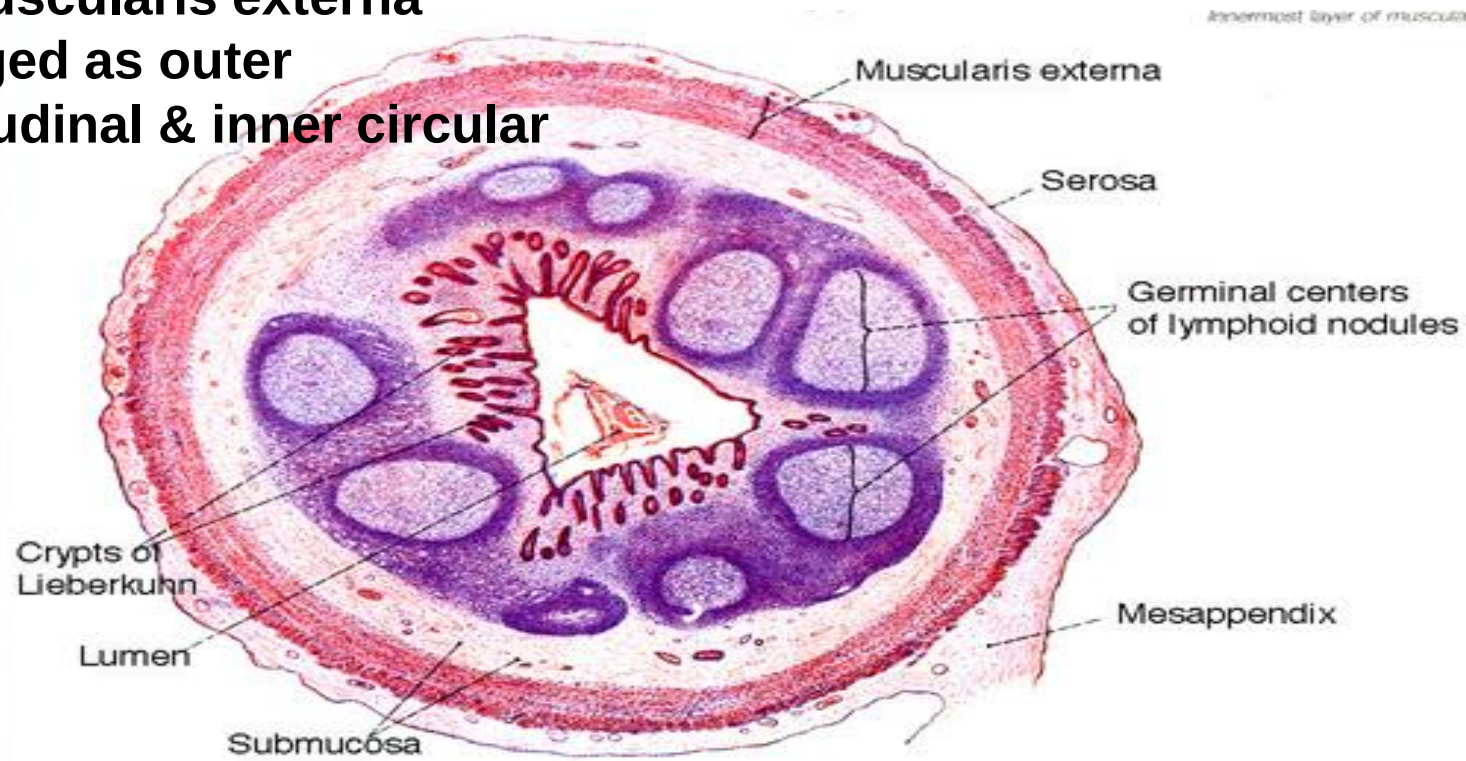
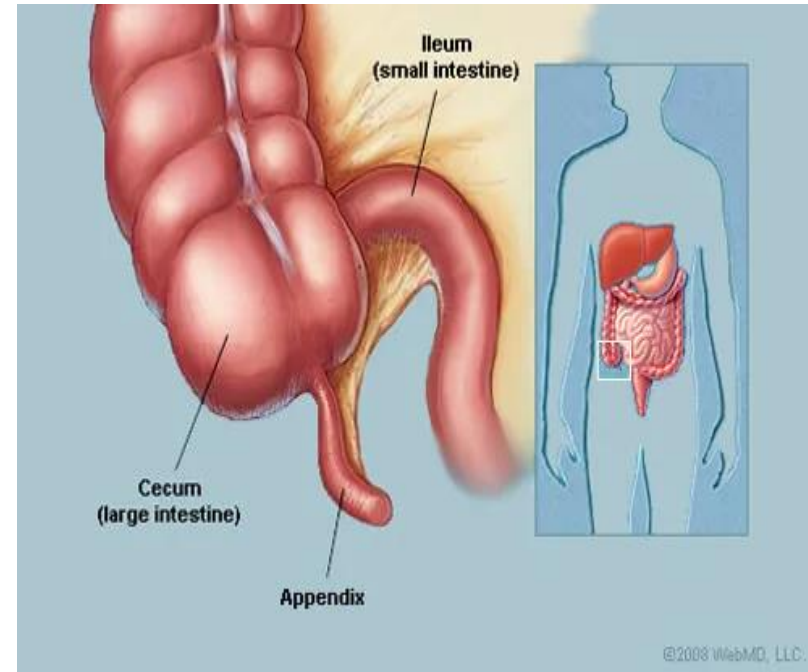


- In the submucosa of the anal canal there is prominent venous plexus.
- The internal hemorrhoidal plexus located above the level of pectinate line, while the external hemorrhoidal plexus lies near the lower end of the canal.
- The anal canal is surrounded by circular & longitudinal layer of muscle continuous with those of the rectum.
- The circular muscle is thickened to form the internal anal sphincter, which is smooth muscle & its under autonomic control.
- Outside the layer of muscle is the external anal sphincter which is made up of striated muscle & its under the voluntary control.



Appendix: Is small blind ending
is evagination of cecum
Long -5-10cm. Diameter-0.8
cm , has small , narrow,
irregular lumen & abundant
lymphatic nodules in it is
wall. Similar to colon but -
fewer , shorter glands
-has no teniae coli

the muscularis externa
arranged as outer
longitudinal & inner circular



THANKS